BAO CAO KIEM THU TONG KET CUOI KY (FINAL TEST SUMMARY REPORT)
CHUNG CHI DAM BAO CHAT LUONG PHAN MEM (SOFTWARE QUALITY ASSURANCE SIGN-OFF)
Du an: EVManager - He thong Quan ly Dich vu Su kien & Tiec cuoi (LV34-001)
Hoc phan: Quan ly Du an CNTT (MAN104) | Cot moc: Tuan 8 - Nghiem thu cuoi ky
Nguoi thuc hien (QA Lead): Bui Nguyen Chi Hau | Phe duyet (PM): Phan Le Huy Hoang
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1. TONG HOP KET QUA KIEM THU TU TUAN 1 DEN TUAN 8:
- Tong so kich ban kiem thu thiet ke: 132 kich ban (100% bao phu)
- So luot thuc thi kiem thu (Test Runs): 296 luot chay xuyen suot 8 tuan
- Kiem thu tu dong Backend (Unit & Integration): 82/82 Tests Passed in 17.8s (100%)
- Ket qua kiem thu he thong (System Test): 35/35 Test Cases Passed (100%)
- Ket qua kiem thu UAT (Nguoi dung doanh nghiep): 15/15 Scenarios Accepted (100%)
- Ket qua kiem thu hieu nang (Performance 50 VUs): p95 = 245ms (< 500ms SLA), Error = 0.00%
- Tong so loi phat hien va khac phuc: 10/10 Defects Closed (BUG-01 -> BUG-08, BUG-104, BUG-105)
- So loi con ton dong: 0 loi (Zero Critical / Blocker / High Open Bugs)

2. MA TRAN BAO PHU YEU CAU (TRACEABILITY MATRIX - 100% USE CASES):
- UC-01 (Dang nhap): TC_AUTH_01..03, LoginNormalizationRegressionTest, UAT-01 [PASS 100%]
- UC-02 (Phan quyen RBAC): TC_AUTH_04..06, UserServiceTest, UAT-02 [PASS 100%]
- UC-03 (Quan ly Khach hang): TC_CUST_01..03, UAT-03, UAT-04 [PASS 100%]
- UC-04 (Tra cuu sanh): TC_VENUE_01..02, VenueServiceTest (9 tests), UAT-05, UAT-06 [PASS 100%]
- UC-05 (Quan ly sanh & giu cho): TC_VENUE_03..06 [PASS 100%]
- UC-06 (Thuc don & mon an): TC_MENU_01..04, MenuServiceTest (3 tests), UAT-07 [PASS 100%]
- UC-07 (Dich vu di kem): TC_MENU_05..08, UAT-08 [PASS 100%]
- UC-08 (Tinh chi phi & bao gia): TC_EVENT_04..06, ContractPricingRegressionTest, UAT-09 [PASS 100%]
- UC-09 (Lap lich su kien): TC_EVENT_01..02, TimezoneSyncRegressionTest, UAT-10, UAT-11 [PASS 100%]
- UC-10 (Xung dot lich sanh): TC_EVENT_03, EventConcurrencyTest, ConflictEdgeCases, UAT-12 [PASS 100%]
- UC-11 (Quan ly hop dong): TC_EVENT_07..08, ContractServiceTest, UAT-13 [PASS 100%]
- UC-12 (Quan ly thanh toan & coc): TC_EVENT_09, TC_AUTH_06, UAT-13 [PASS 100%]
- UC-13 (Dieu phoi sanh & nhan su): UAT-08, UAT-11, TC_EVENT_01 [PASS 100%]
- UC-14 (Dashboard & bao cao): UAT-14, UAT-15, TC_AUTH_01 [PASS 100%]
=> 14/14 Use Cases dat chuan bao phu (Moi UC co tu 3-14 Test Cases kiem chung)

3. DANH GIA MUC DO SAN SANG TRIEN KHAI (PRODUCTION READINESS):
- Do hoan thien chuc nang: 100% hoan tat 3 luong nghiep vu loi
- Hieu nang & Tai: Dat vuot muc SLA (50 VUs on dinh, p95=245ms, PageLoad < 2.2s)
- Do tin cay CSDL: 15 bang chuan 3NF (schema.sql), Pessimistic Write Lock chong race condition
- Bao mat: BCrypt hash, JWT stateless token, RBAC 4 vai tro chat che o Backend
- Kha nang phuc hoi: 16 ban Flyway Migration (V1-V16) dong bo tu dong, san sang Docker Staging

4. BIEN BAN NGHIEM THU CHAT LUONG (FINAL QUALITY SIGN-OFF):
- Dai dien Ban Kiem thu (QA Lead): Bui Nguyen Chi Hau (Da ky xac nhan)
- Dai dien Ban Quan ly Du an (PM): Phan Le Huy Hoang (Da phe duyet nghiem thu)
- Ngay ky: 18/10/2026 | Ket luan: HE THONG EVMANAGER DU DIEU KIEN DONG DU AN & BAO VE
- Chi tiet xem tai: docs/testing/final-test-report.md